

Soutenance du TER

Rendu de pelage et plumage

Auteurs

Arnaud Peler - Algo

Clément Potier - Imagine

Isaac Lluís - Imagine

Ryan Rodrigues - Imagine

Huppe Huppe Hourra

Encadrant

Nicolas Lutz

Rapporteur

Marc Hartley



UNIVERSITÉ DE
MONTPELLIER



FACULTÉ DES SCIENCES
DE MONTPELLIER

Sommaire

- ▶ Introduction
- ▶ Prérequis
- ▶ PBR
- ▶ Pelage
- ▶ Plumage
- ▶ Rendu
- ▶ Conclusion



Introduction

Introduction



Huppe fasciée¹



Plume²



Cheveux³

¹ <https://i.pinimg.com/originals/cf/59/99/cf5999ac0a7ca4ca1acf584f4ae6d011.jpg>

² <https://cdn.create.vista.com/api/media/small/20214329/stock-photo-buzzard-feather>

³ <https://static.vecteezy.com/system/resources/thumbnails/071/009/622/small/close-up-view-of-long-hair-illuminated-by-soft-light-in-natural-setting-free-photo.jpeg>

- ▶ **Objectif: Faire un rendu de cheveux**
- ▶ **Objectif : Retranscrire et vulgariser les articles (Marschner et al. 2003), (d' Eon et al. 2011), (Chiang et al. 2016) et (Padrón-Griffe et al. 2024)**

Introduction



(d' Eon et al. 2011)

Introduction



(Chiang et al. 2016)

Introduction

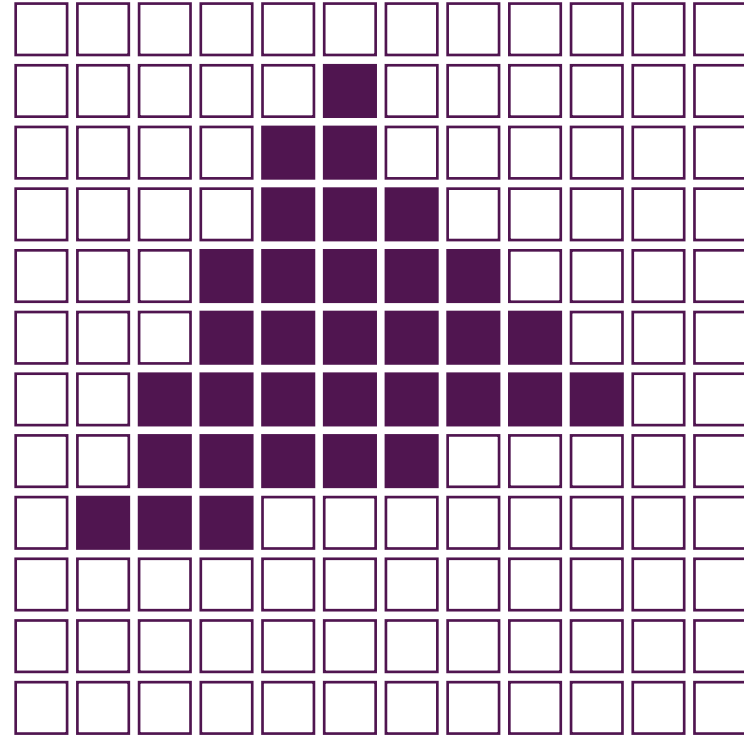
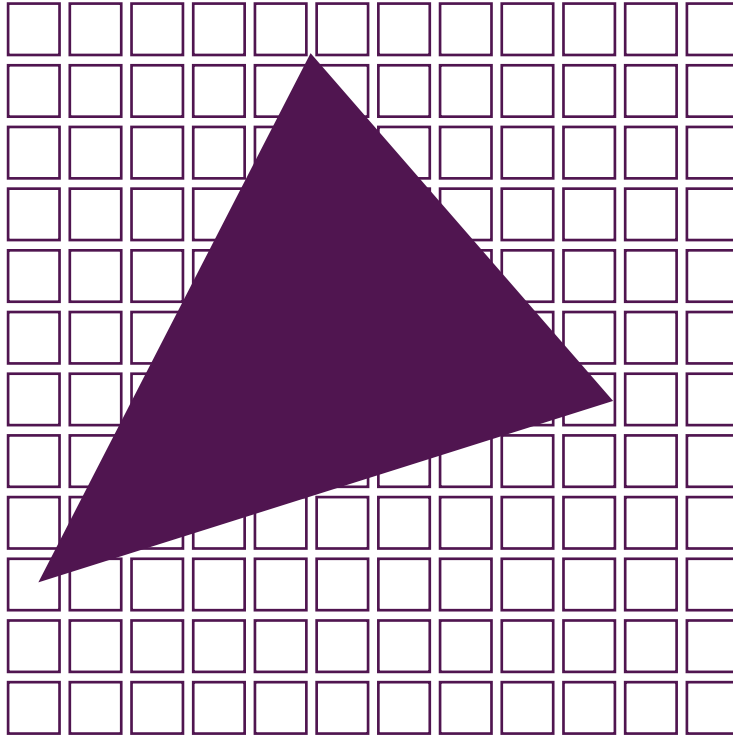


(Padrón-Griffe et al. 2024)



Prérequis

Rastérisation



- ▶ Déterminer la couleur d'un pixel
- ▶ À partir de la géométrie locale et d'éclairage
- ▶ Bidirectional Scattering Distribution Function
- ▶ RGB

- ▶ Luminosité
- ▶ Sur une surface
- ▶ Avec une direction
- ▶ Noté avec $L(p, \vec{\omega})$

A large, stylized purple leaf graphic with white veins, positioned behind a dark purple rectangular box. The leaf is oriented vertically, with its base at the bottom and its tip at the top.

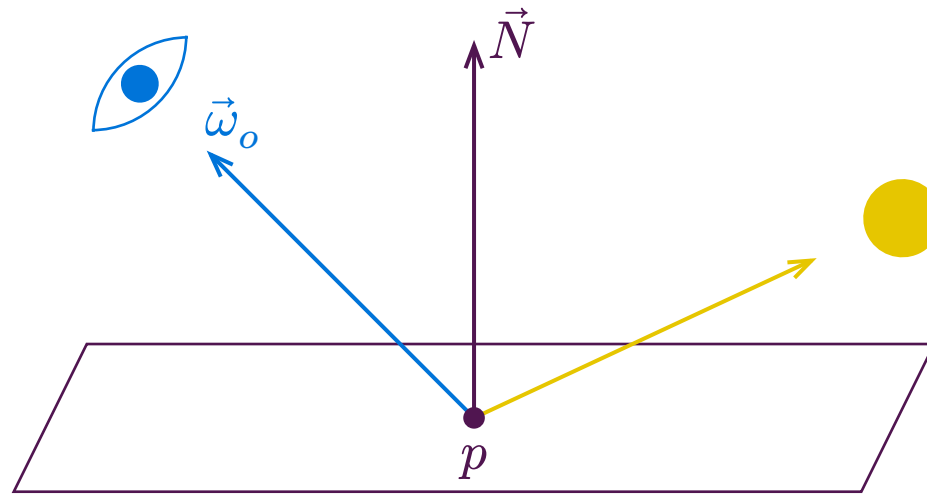
PBR

- ▶ Imitation de modèles physiques de lumière
- ▶ Conservation d'énergie
- ▶ Prise en compte de micro-structures
- ▶ Fondé sur des mesures physiques

$$L_o(p, \vec{\omega}_o) = \int_{\Omega} f_r(p, \vec{\omega}_i, \vec{\omega}_o) \times L_i(p, \vec{\omega}_i) \times (\vec{N} \cdot \vec{\omega}_i) d\vec{\omega}_i$$

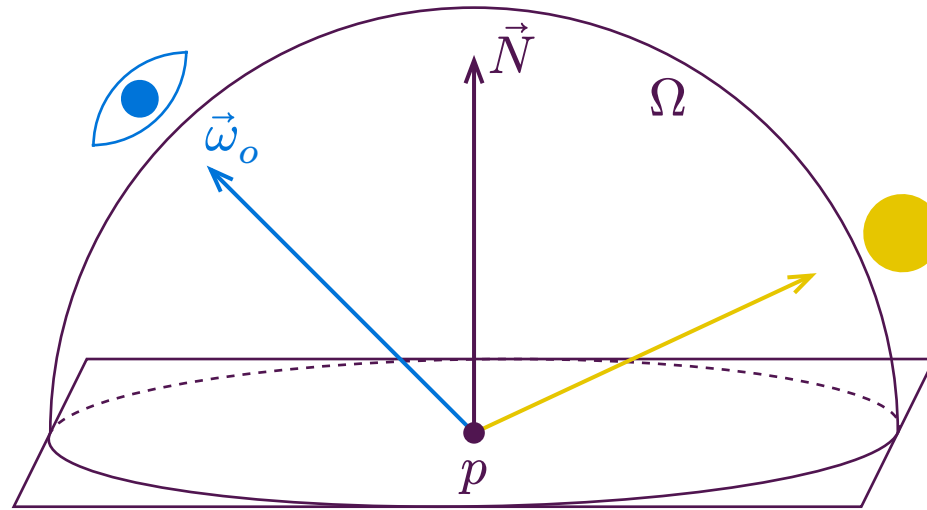
Équation de rendu

$$L_o(p, \vec{\omega}_o) = \int_{\Omega} f_r(p, \vec{\omega}_i, \vec{\omega}_o) \times L_i(p, \vec{\omega}_i) \times (\vec{N} \cdot \vec{\omega}_i) d\vec{\omega}_i$$



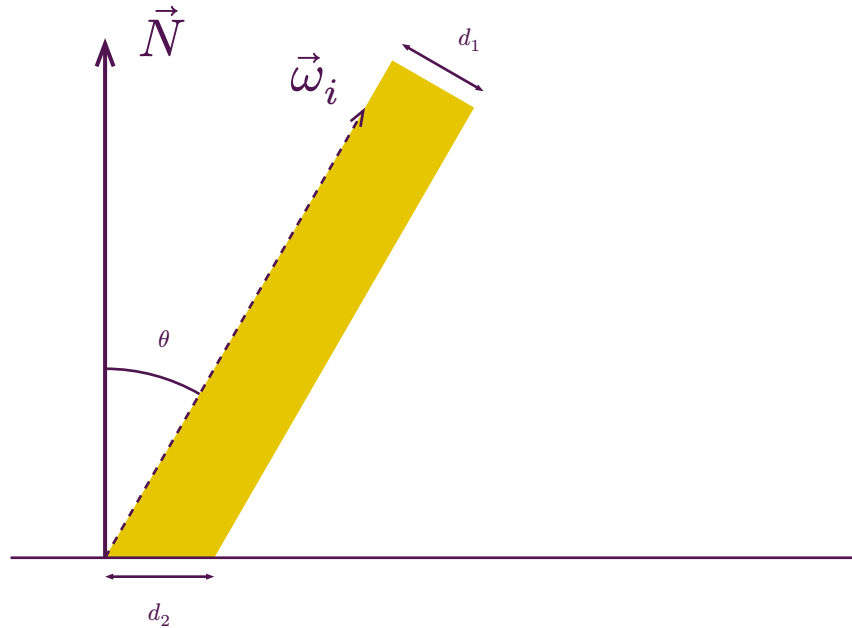
Équation de rendu

$$L_o(p, \vec{\omega}_o) = \int_{\Omega} f_r(p, \vec{\omega}_i, \vec{\omega}_o) \times L_i(p, \vec{\omega}_i) \times (\vec{N} \cdot \vec{\omega}_i) d\vec{\omega}_i$$

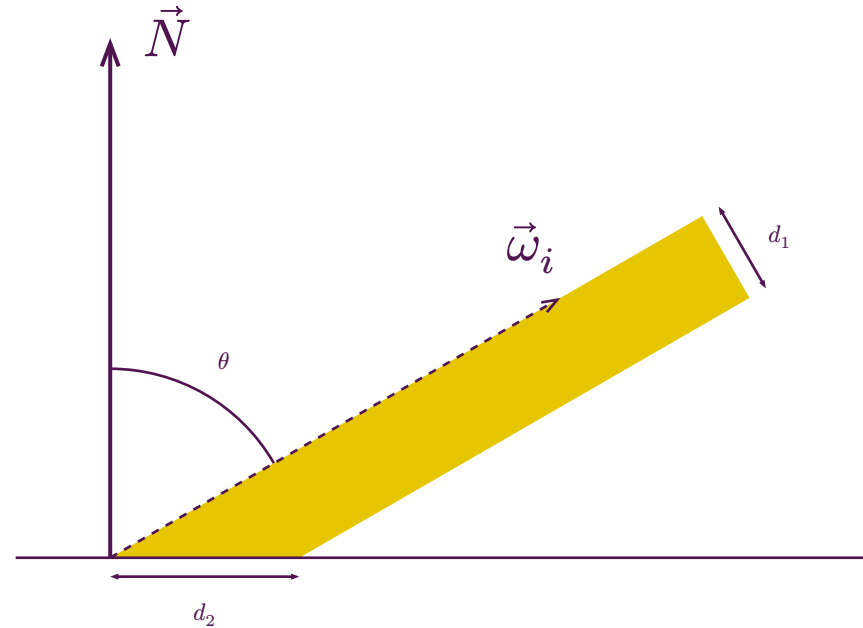


Équation de rendu

$$L_o(p, \vec{\omega}_o) = \int_{\Omega} f_r(p, \vec{\omega}_i, \vec{\omega}_o) \times L_i(p, \vec{\omega}_i) \times \left(\vec{N} \cdot \vec{\omega}_i \right) d\vec{\omega}_i$$



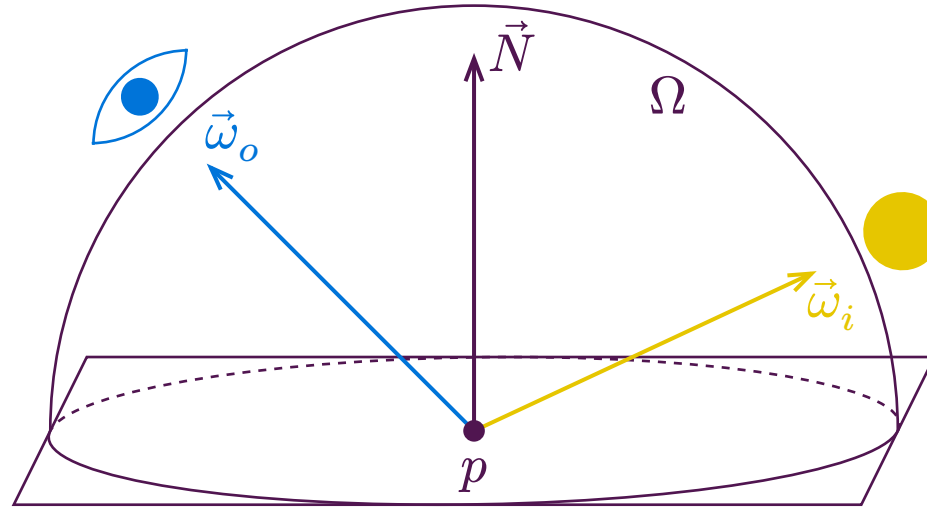
$$\theta = 30^\circ, d_2 = 0.58$$



$$\theta = 60^\circ, d_2 = 1$$

Équation de rendu

$$L_o(p, \vec{\omega}_o) = \int_{\Omega} f_r(p, \vec{\omega}_i, \vec{\omega}_o) \times L_i(p, \vec{\omega}_i) \times (\vec{N} \cdot \vec{\omega}_i) d\vec{\omega}_i$$

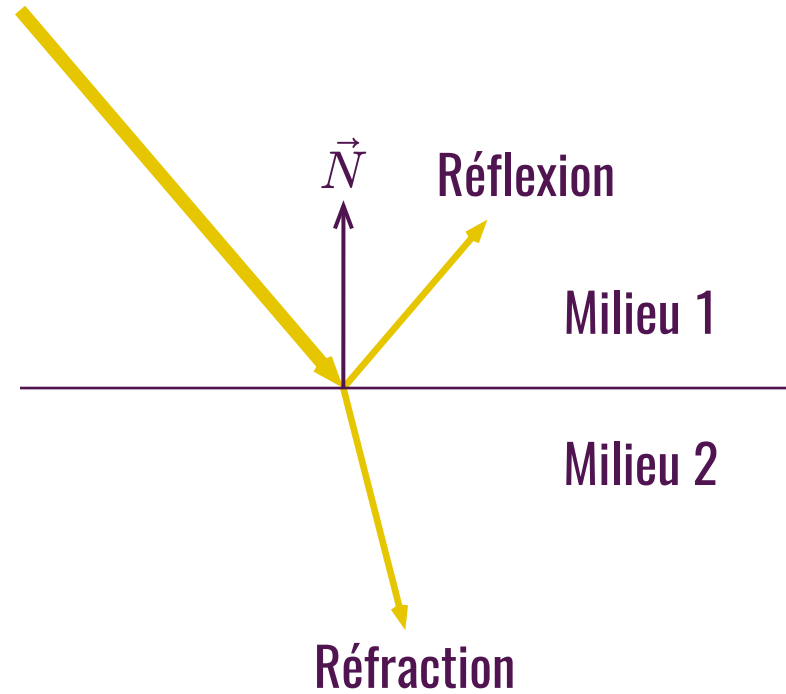


$$L_o(p, \vec{\omega}_o) = \int_{\Omega} f_r(p, \vec{\omega}_i, \vec{\omega}_o) \times L_i(p, \vec{\omega}_i) \times (\vec{N} \cdot \vec{\omega}_i) d\vec{\omega}_i$$

- BRDF qui dépends de la méthode utilisée

Équation de rendu

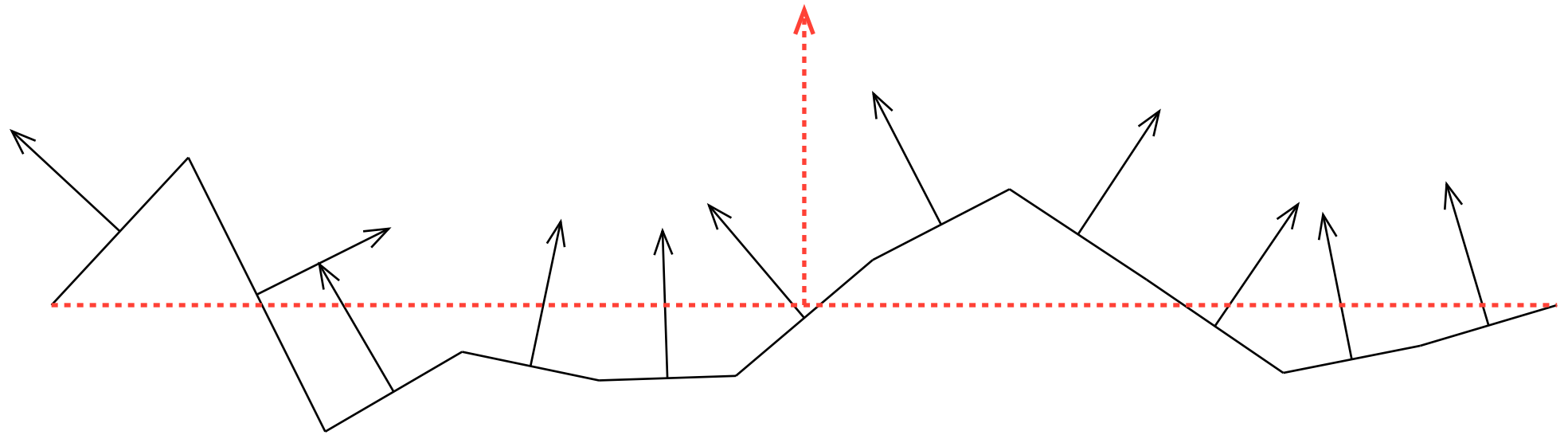
$$L_o(p, \vec{\omega}_o) = \int_{\Omega} f_r(p, \vec{\omega}_i, \vec{\omega}_o) \times L_i(p, \vec{\omega}_i) \times (\vec{N} \cdot \vec{\omega}_i) d\vec{\omega}_i$$



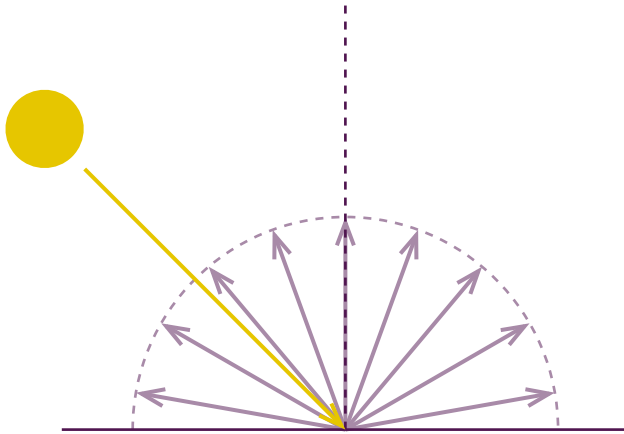
Modèle Cook-Torrance

- ▶ Modèle PBR très utilisé (Unreal Engine)
- ▶ Capable de temps réel
- ▶ Prends en compte: micro-facettes, métallicité, occlusion ambiante

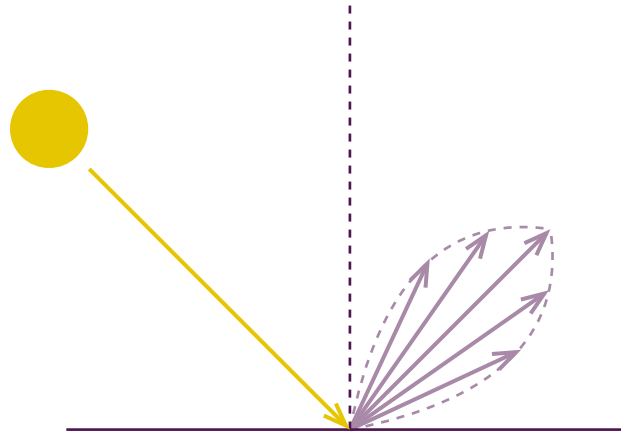
Micro-facettes



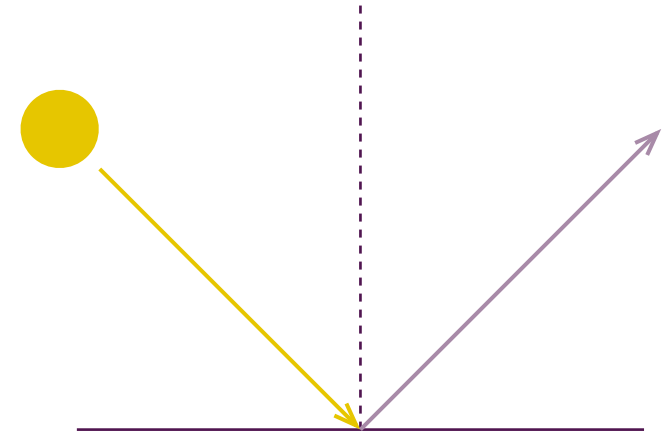
Rugosité



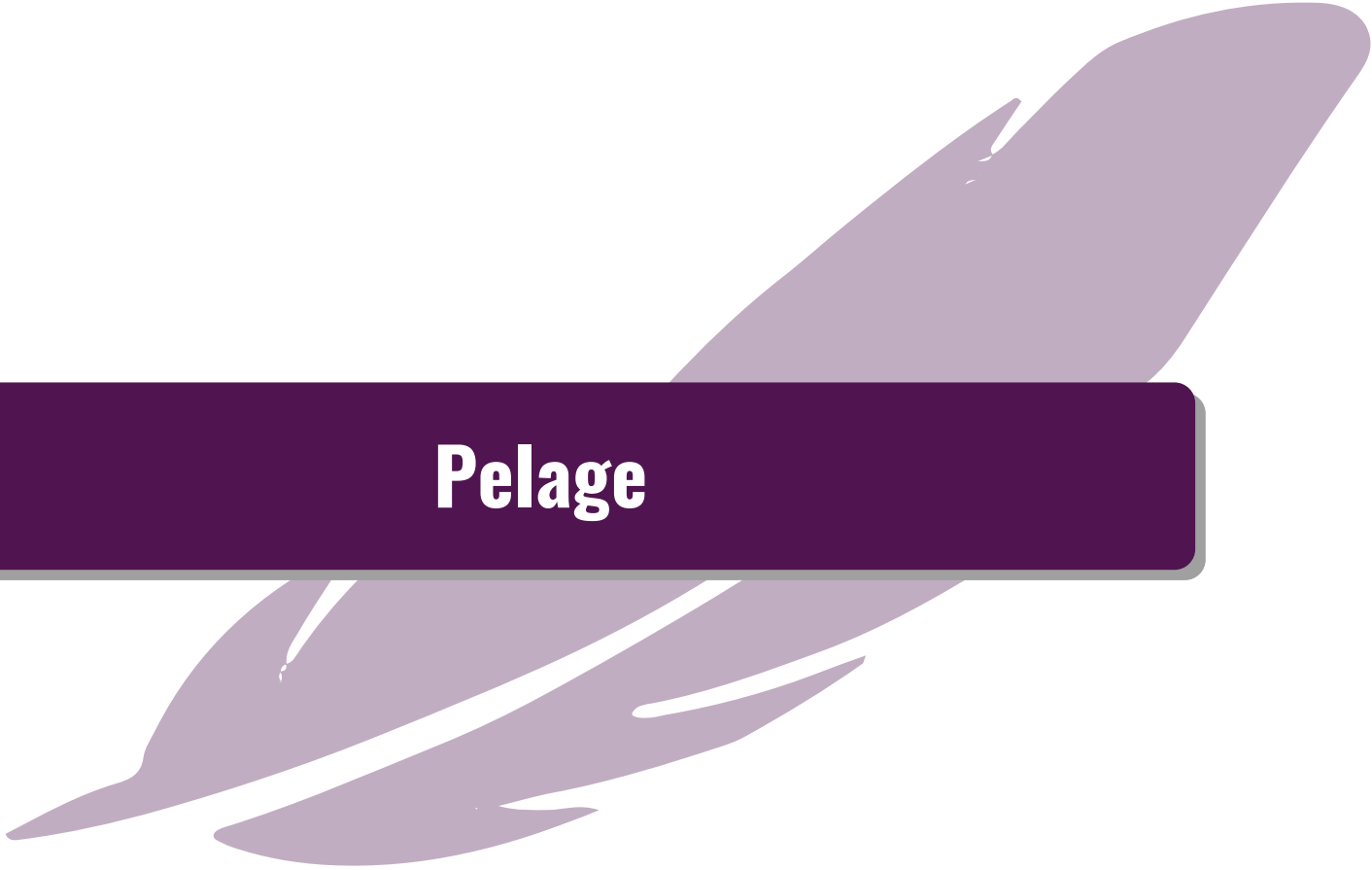
Rugosité maximale



Rugosité intermédiaire

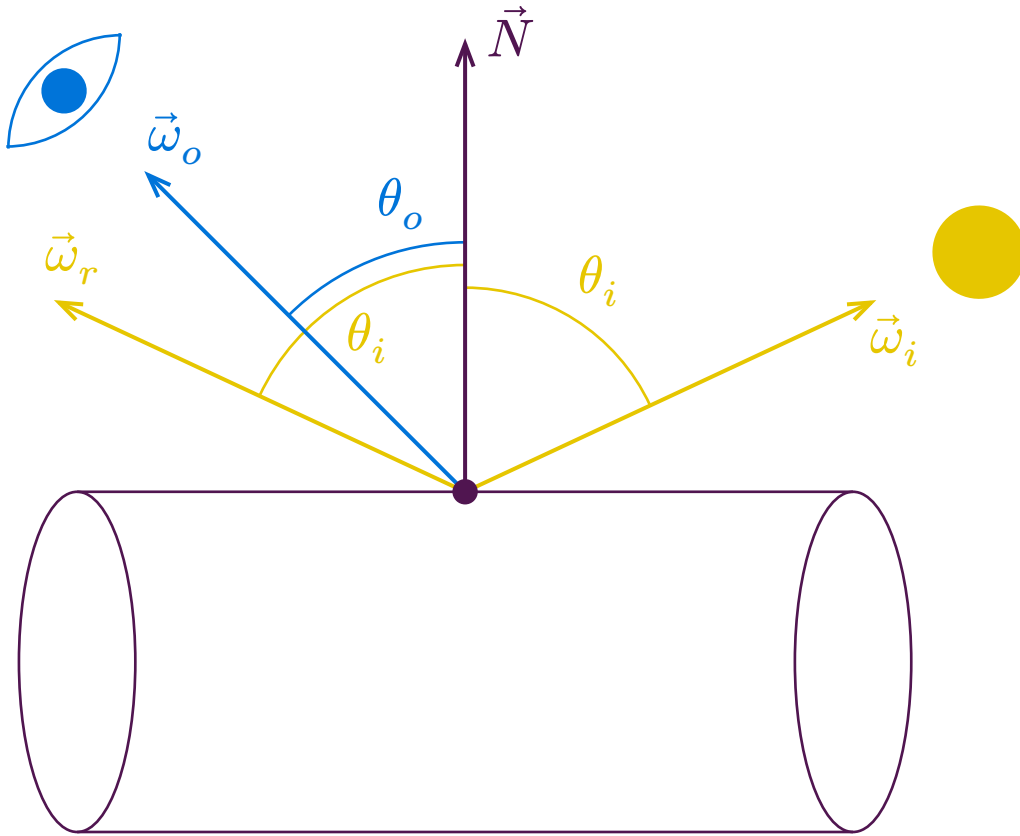


Pas de rugosité

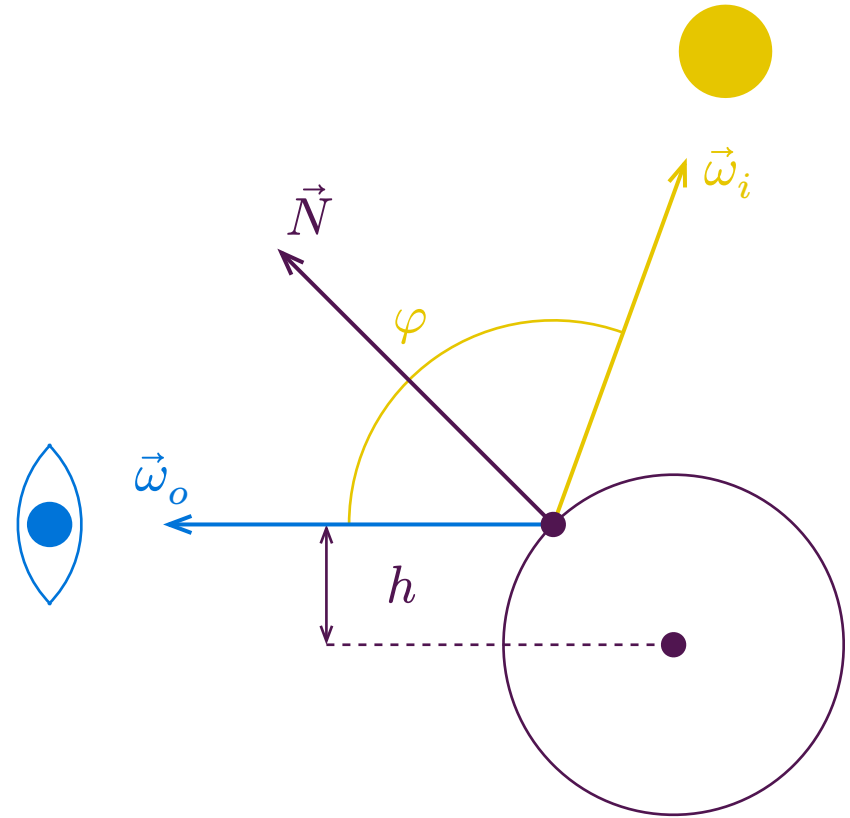


Pelage

Modélisation d'un cheveu

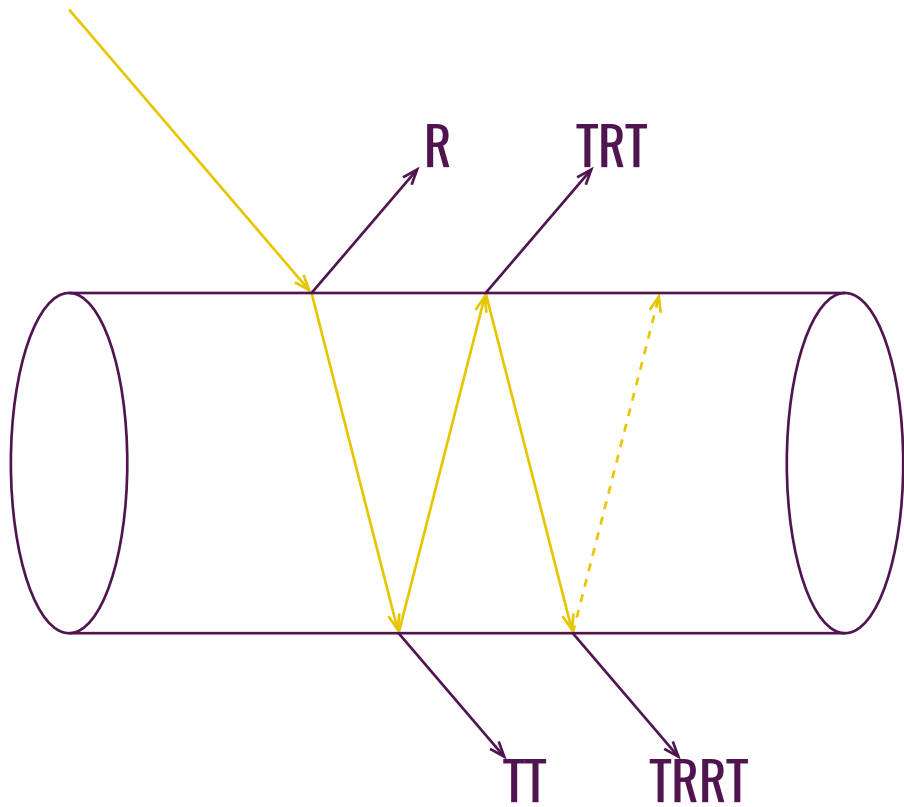


Vue longitudinale

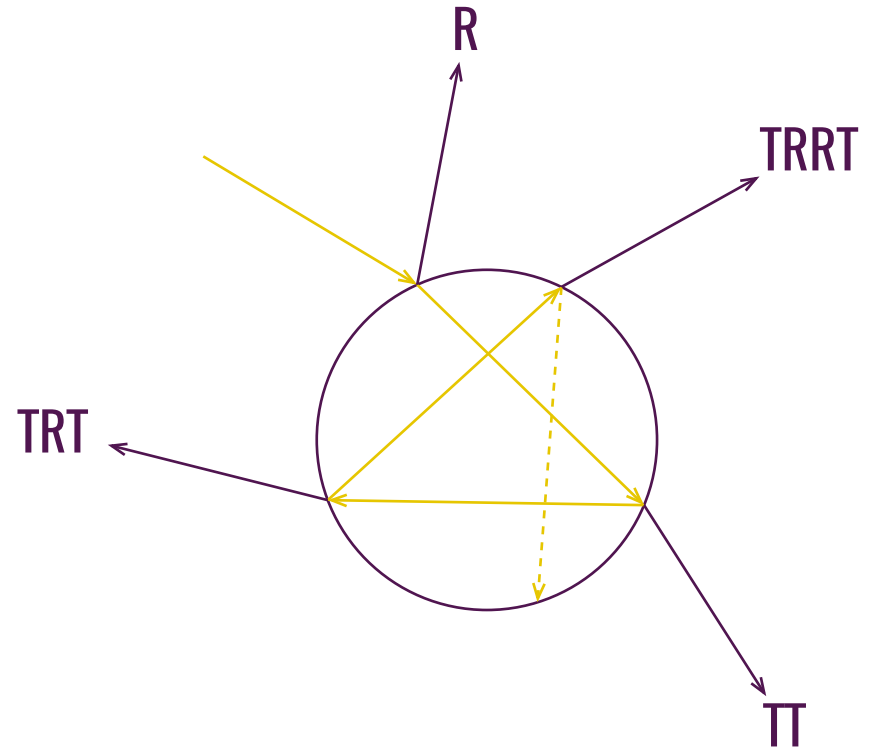


Vue azimutale

La lumière au sein d'un cheveu

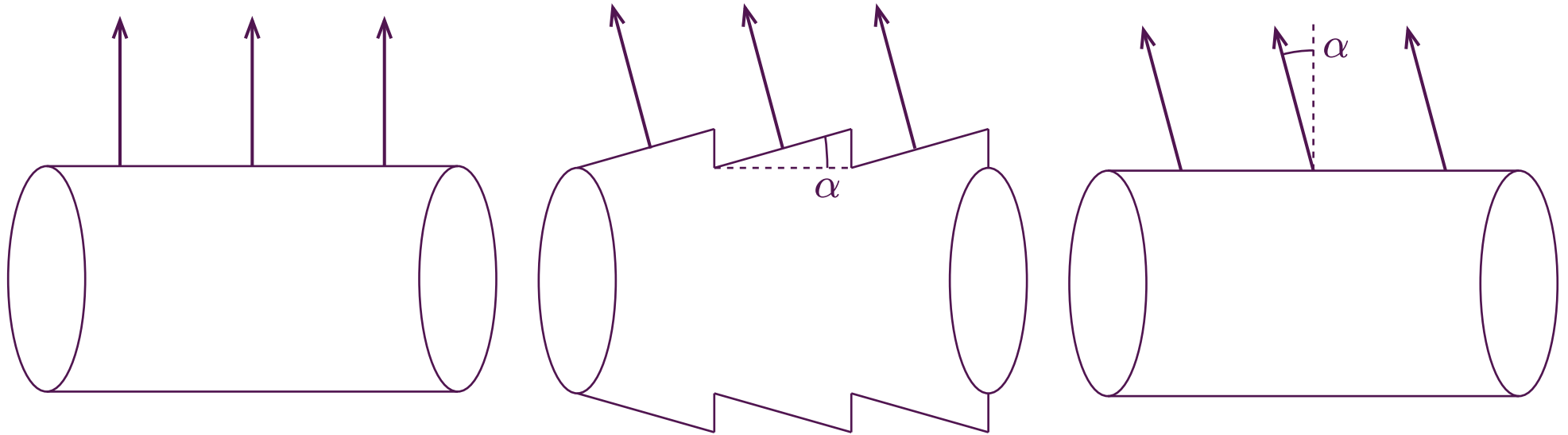


Vue longitudinale



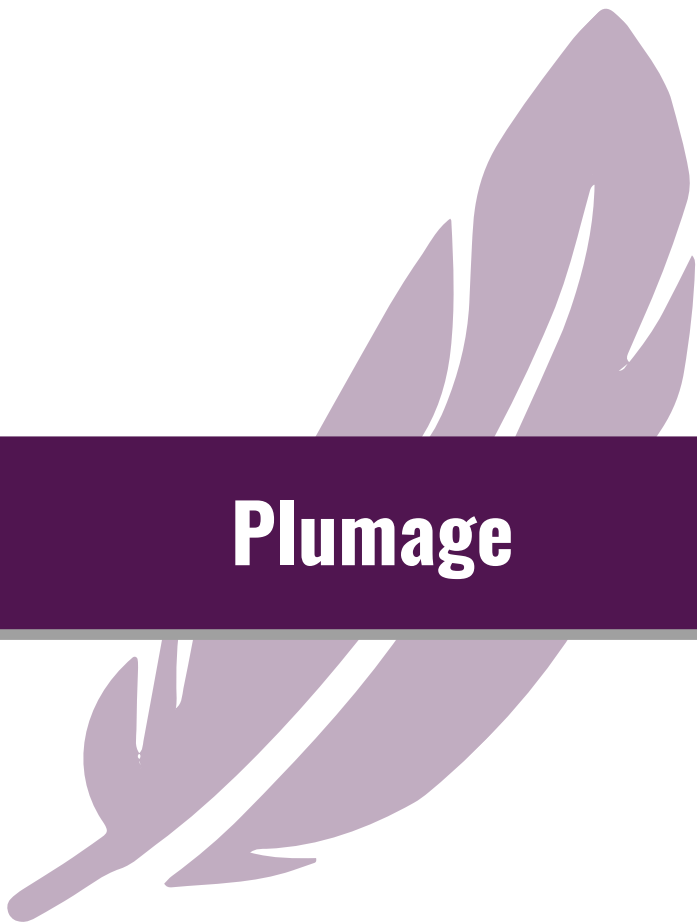
Vue azimutale

Les cuticules



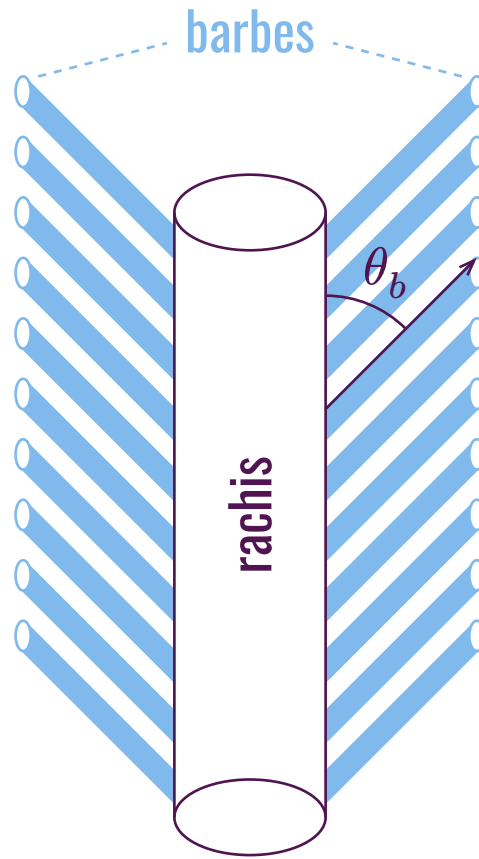
Paramètres supplémentaires

- ▶ η : Indice de réfraction
- ▶ β_L : Rugosité longitudinale
- ▶ β_A : Rugosité azimutale
- ▶ μ_a : Coefficient d'absorption

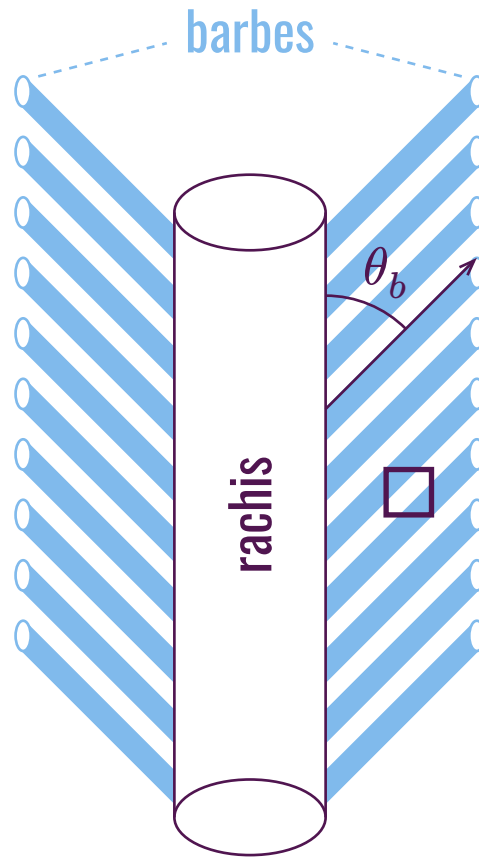


Plumage

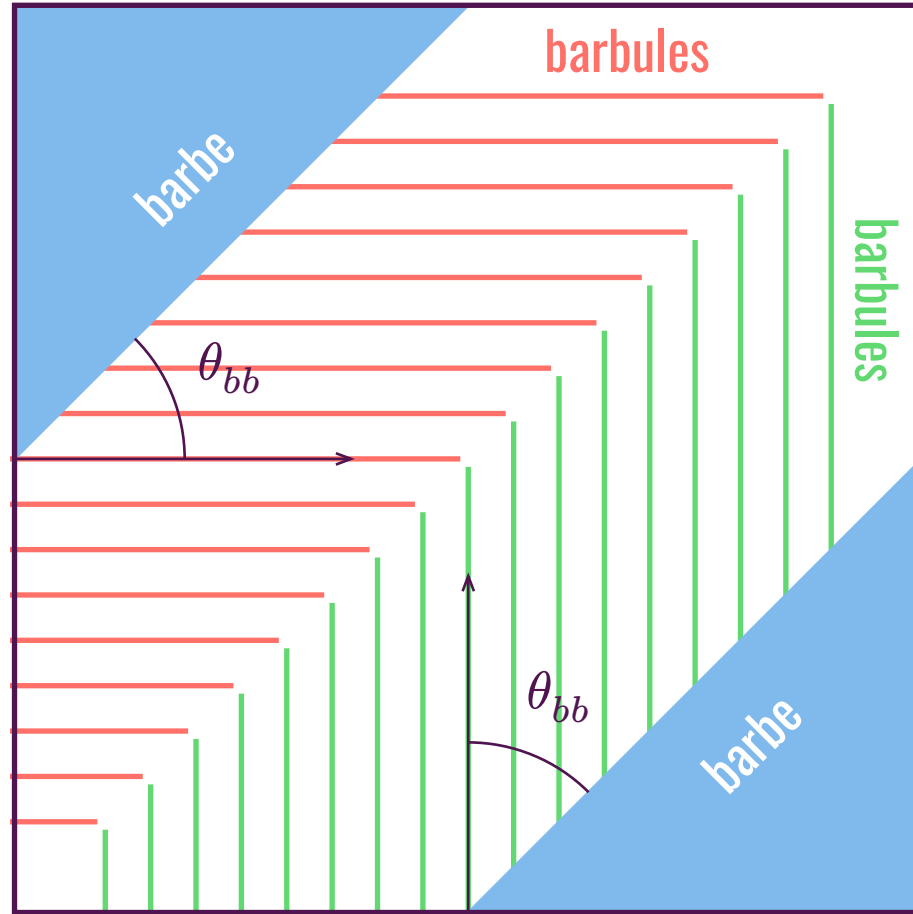
Modélisation d'une plume



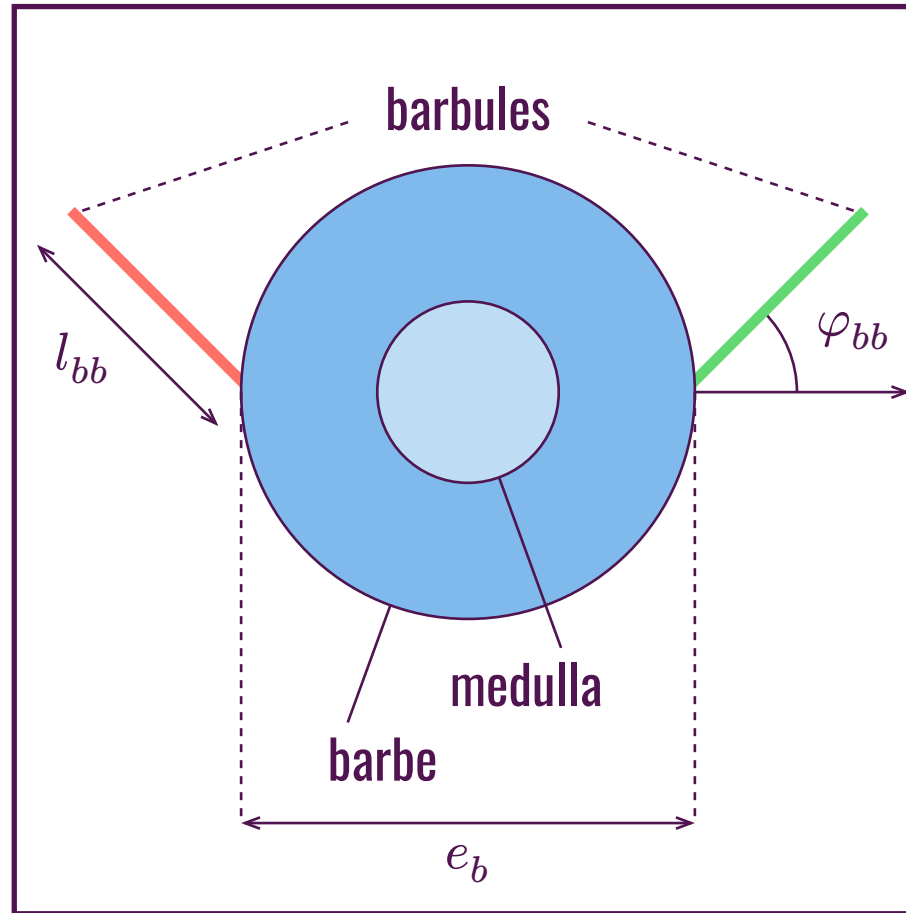
Modélisation d'une plume



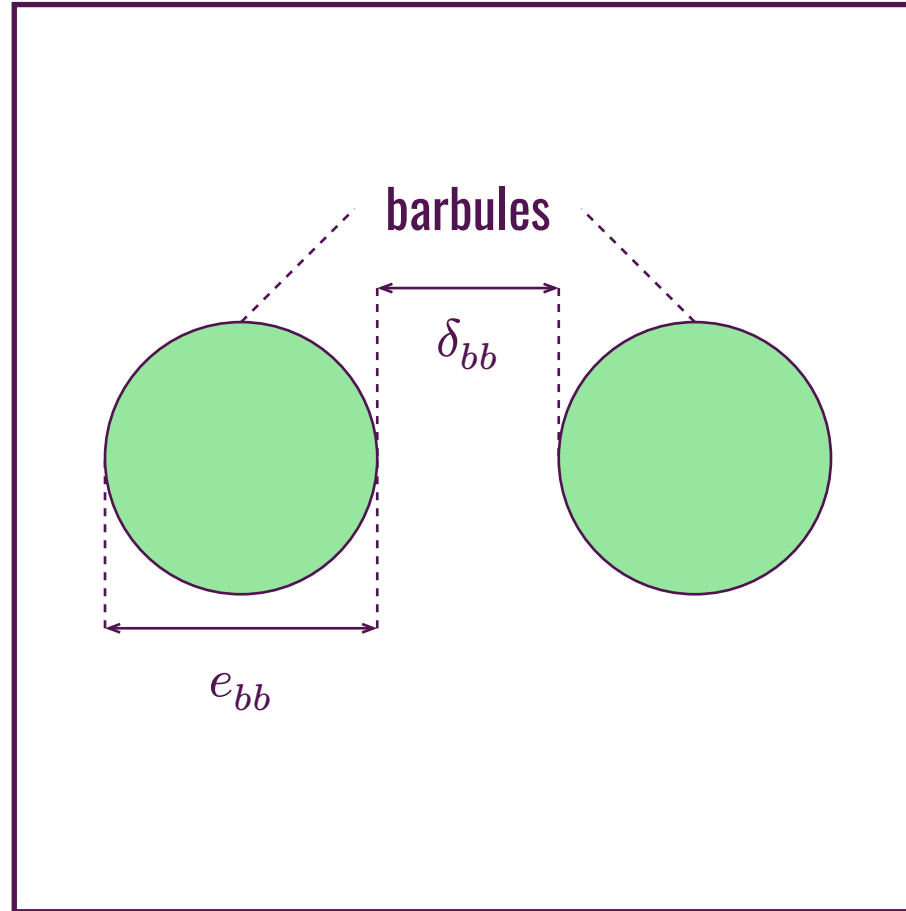
Vue longitudinale entre les barbes



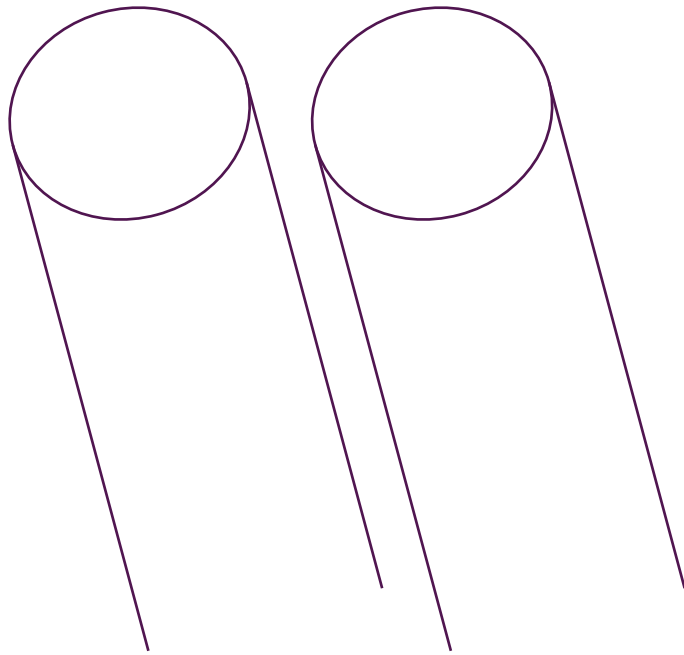
Vue azimutale d'une barbe



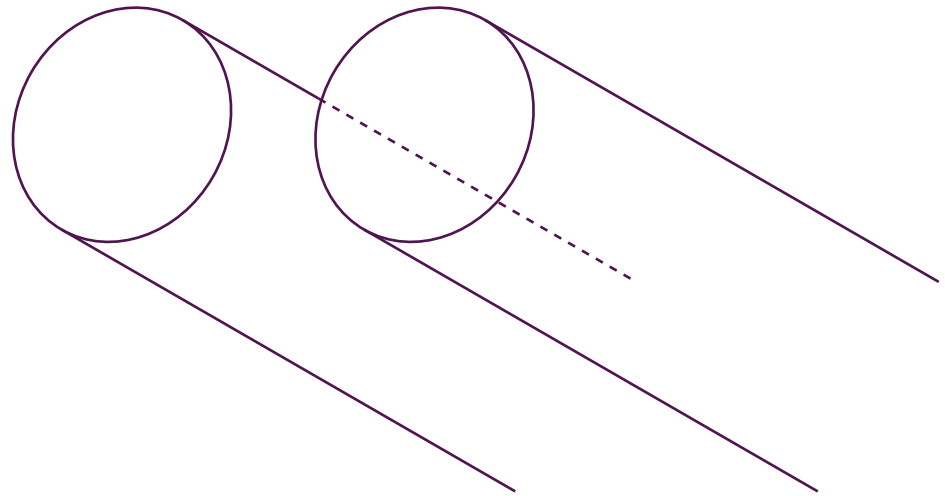
Vue azimutale des barbules



Masquage



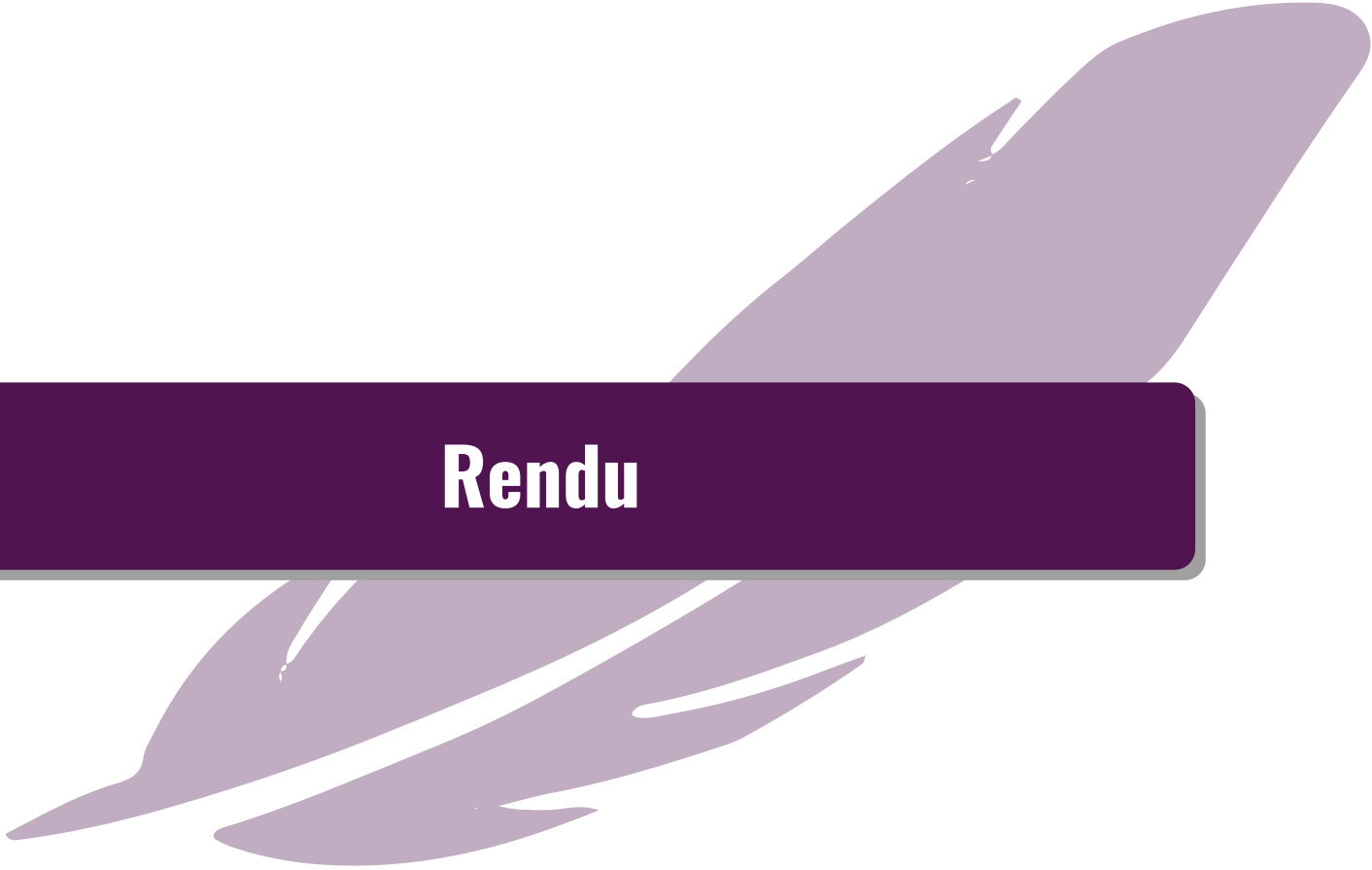
75°



30°

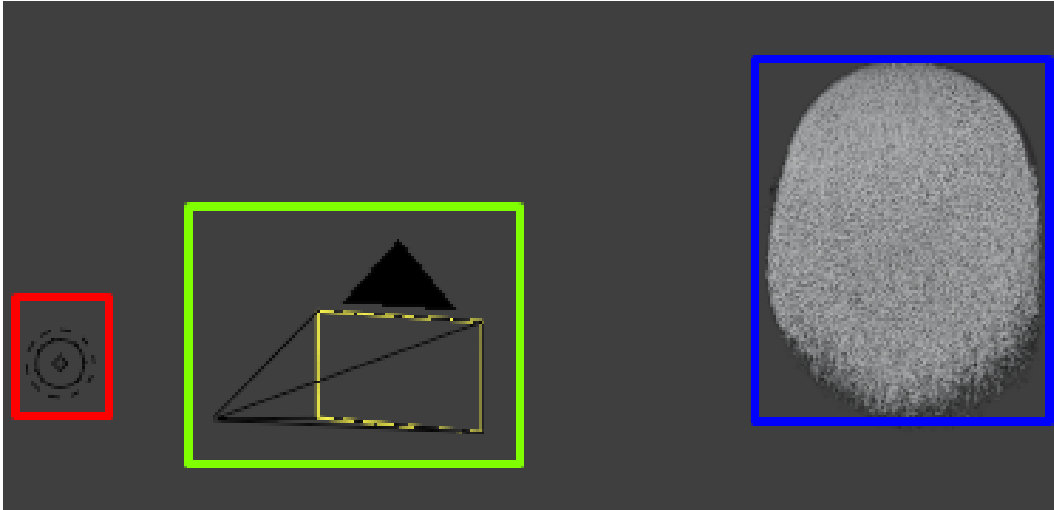
Paramètres supplémentaires

- ▶ Paramètres vus pour les cheveux ($\vec{N}$, $\vec{\omega}_o$, $\vec{\omega}_i$, θ_o , θ_i , h)
- ▶ β_m : Rugosité longitudinale de la barbe
- ▶ β_n : Rugosité azimutale de la barbe
- ▶ a_b : Rayon de la barbe
- ▶ $\sigma_{b,a}$: Coefficient d'absorption de la barbe
- ▶ η_b : Indice de réfraction de la barbe
- ▶ a_m : Rayon de la medulla
- ▶ $\sigma_{m,a}$: Coefficient d'absorption de la medulla
- ▶ η_m : Indice de réfraction de la medulla

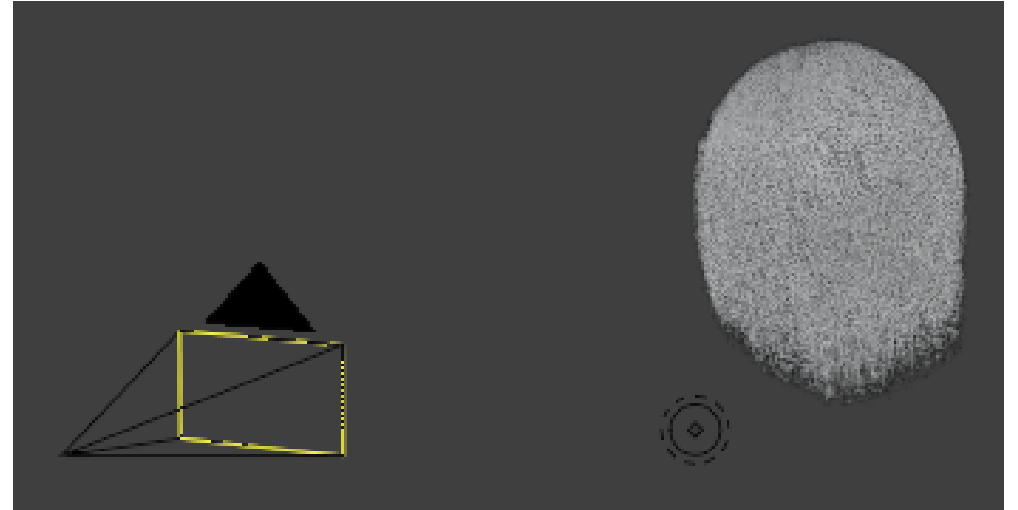
A large, stylized purple brushstroke graphic that sweeps across the middle of the slide, partially obscured by the title bar.

Rendu

Scènes utilisées



Scène « avant »

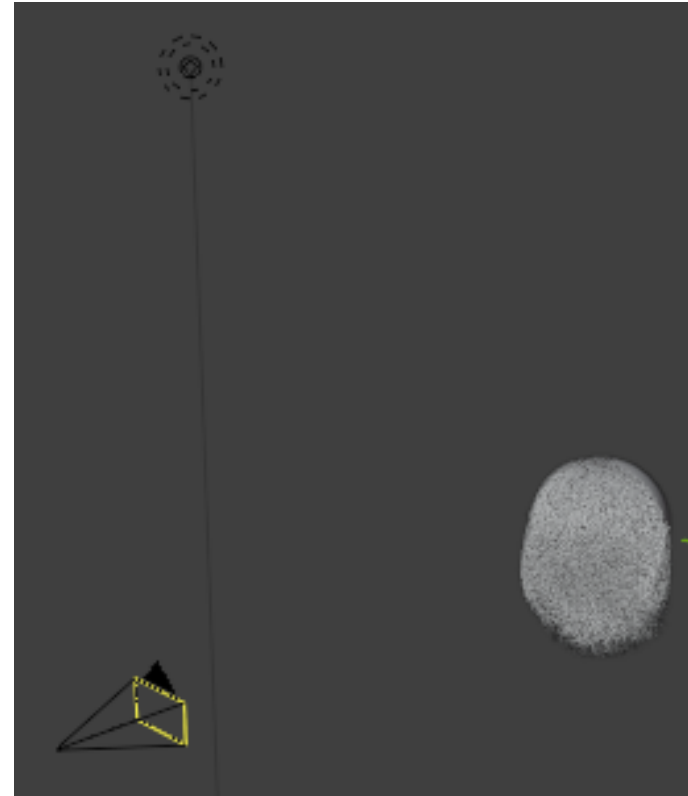


Scène « côté »

Scènes utilisées



Scène « avant-haut »

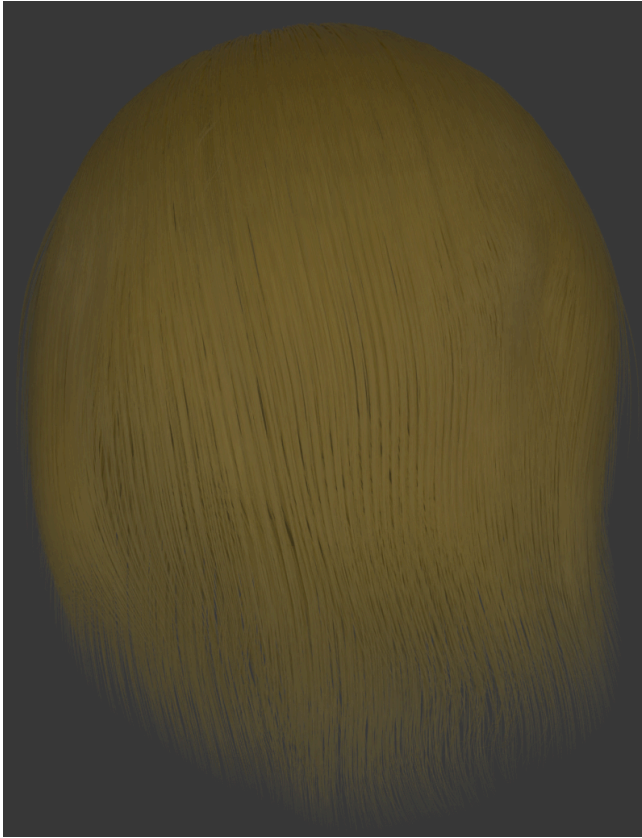


Scène « côté-haut »

Shaders utilisés

- ▶ Blender - (Chiang et al. 2016) (modèle référent)
- ▶ Custom - (Chiang et al. 2016)
- ▶ Custom - (d' Eon et al. 2011)

Rendu de la scène « avant »



Blender - (Chiang et al. 2016)



Custom - (Chiang et al. 2016)

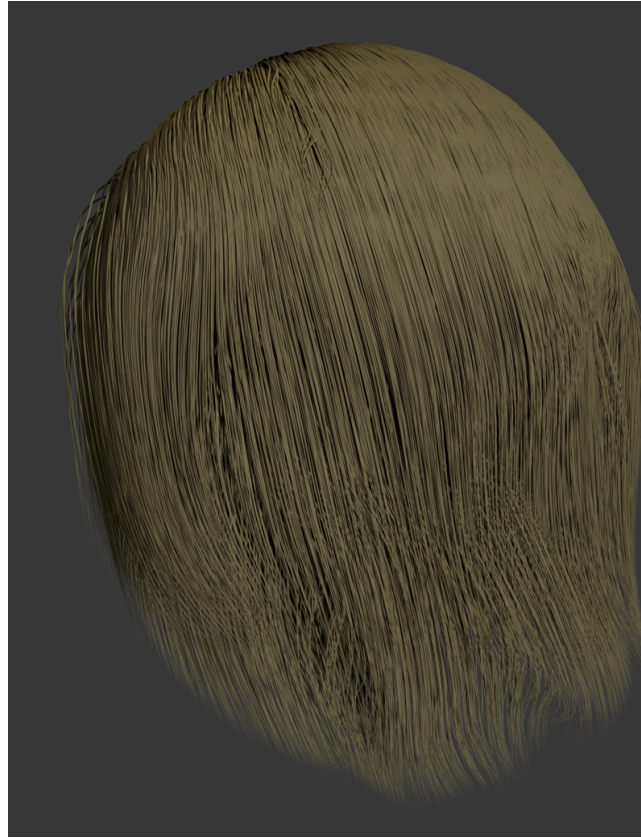


Custom - (d'Eon et al. 2011)

Rendu de la scène « côté »



Blender - (Chiang et al. 2016)



Custom - (Chiang et al. 2016)



Custom - (d'Eon et al. 2011)

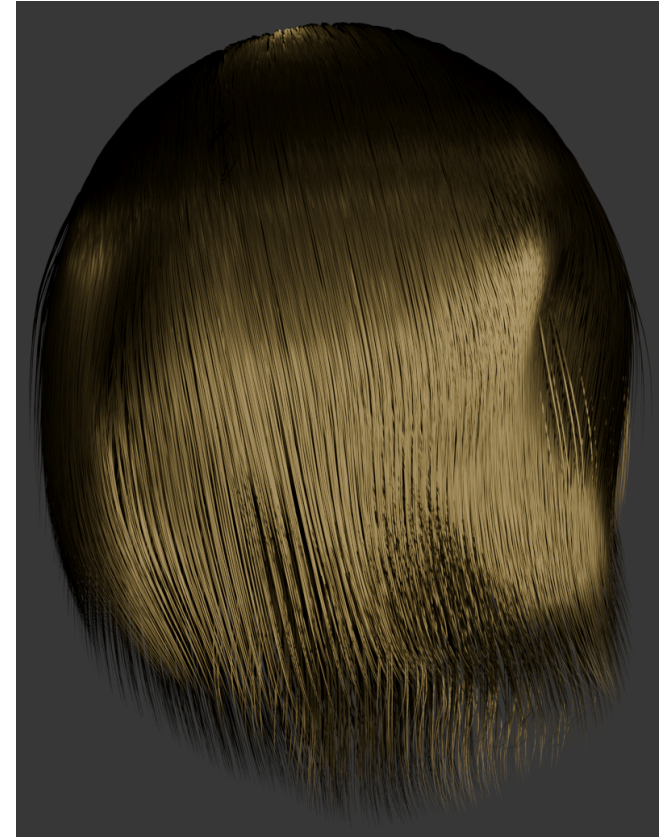
Rendu de la scène « avant-hauteur »



Blender - (Chiang et al. 2016)

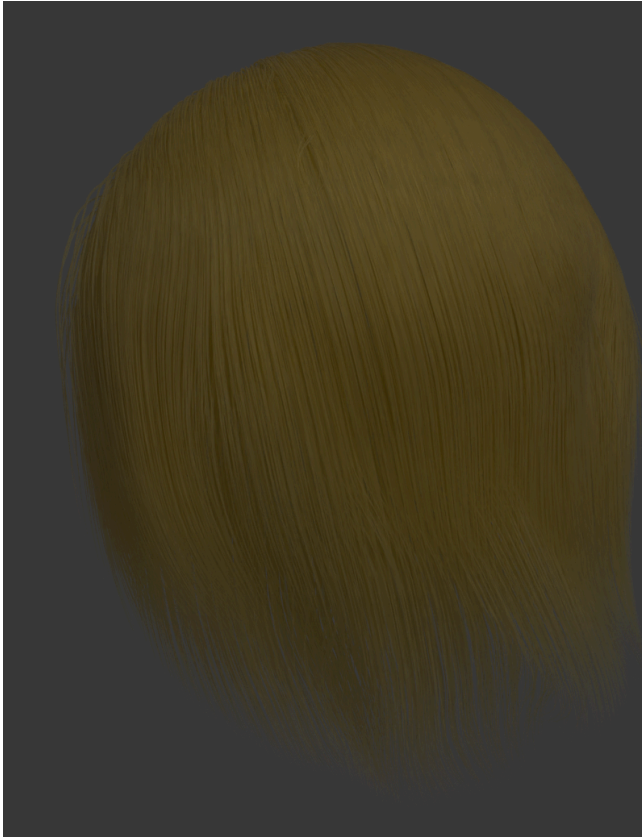


Custom - (Chiang et al. 2016)



Custom - (d'Eon et al. 2011)

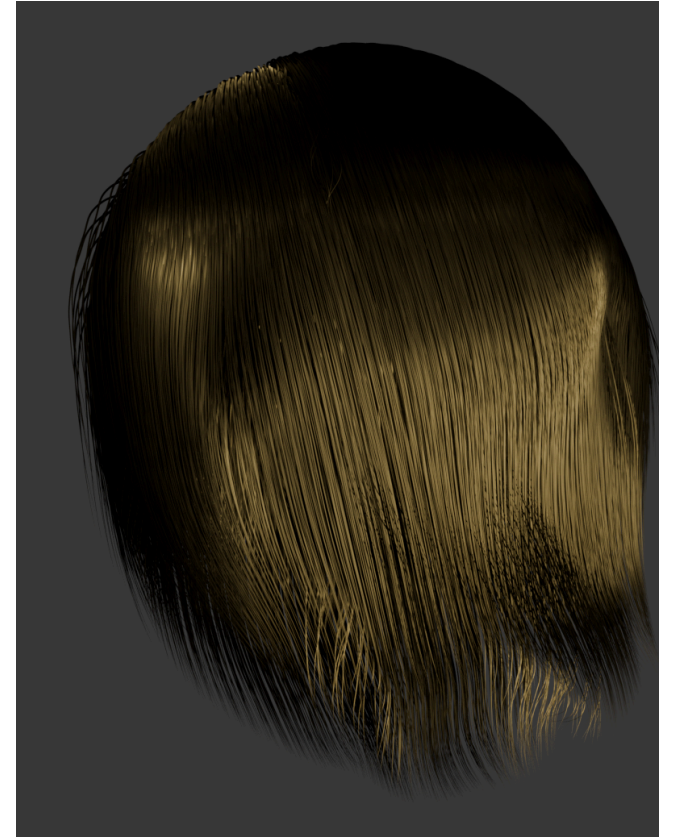
Rendu de la scène « côté-hauteur »



Blender - (Chiang et al. 2016)



Custom - (Chiang et al. 2016)



Custom - (d'Eon et al. 2011)

La machine utilisée

- ▶ Logiciel : Blender 5.1
- ▶ Système d'exploitation : Windows 11 Professionnel version 25H2
- ▶ Mémoire vive (RAM) : 32 Go, cadence 3200MT/s
- ▶ Processeur (CPU) : Intel Core i9-11900H, 8 cœurs, 2.50 GHz
- ▶ Carte graphique (GPU) : NVIDIA GeForce RTX 3070, 8 Go de VRAM

Temps de rendu

Shader utilisé : test effectué	Blender - (Chiang et al. 2016)	Custom - (Chiang et al. 2016)	Custom - (d'Eon et al. 2011)
Scène avant	14min 26s 45ms	1min 17s 71ms	1min 41s 90ms
Scène côté	15min 06s 62ms	1min 53s 21ms	1min 43s 90ms
Scène avant-haut	9min 13s 67ms	1m 53s 07ms	4min 32s 86ms
Scène côté-haut	08min 46s 74ms	1min 40s 45ms	3min 05s 56ms

- ▶ Limitation de précision
- ▶ Limitation du moteur de rendu
- ▶ Choix de la couleur pour le papier (d'Eon et al. 2011)



Conclusion

- ▶ Rendu PBR, pelage et plumage
- ▶ Regroupement de plusieurs papiers
- ▶ Découverte de rendus physiquement réalistes

- ▶ Améliorer le rendu
- ▶ Re-essayer de faire le rendu de plumes

Formule de Cook-Torrance

$$L_o(p, \vec{\omega}_o) = \int_{\Omega} f_r(p, \vec{\omega}_i, \vec{\omega}_o) \times L_i(p, \vec{\omega}_i) \times (\vec{N} \cdot \vec{\omega}_i) \quad d\vec{\omega}_i$$

$$f_r(p, \vec{\omega}_i, \vec{\omega}_o) = k_d \times f_d(p) + k_s \times f_s(\vec{\omega}_i, \vec{\omega}_o)$$

$$F = F_0 + (1 - F_0) \times (1 - \cos \theta)^5$$

$$k_s = F$$

$$k_d = 1 - k_s$$

$$F_0 = \text{lerp}(0.04, A, M)$$

$$F_0 = 0.04 * (1 - M) + A * M$$

$$f_s(\vec{\omega}_i, \vec{\omega}_o) = \frac{F \times G_2(\vec{\omega}_o) \times D(\vec{\omega}_i, \vec{\omega}_o)}{4 \cos \theta_i \cos \theta_o}$$

$$G_2(\vec{\omega}_o) = \frac{\vec{N} \cdot \vec{\omega}_o}{(\vec{N} \cdot \vec{\omega}_o) \times (1 - k) + k}$$

$$D(\vec{\omega}_i, \vec{\omega}_o) = \frac{\beta^2}{\pi \times \left(\left(\vec{N} \cdot \frac{\vec{\omega}_i + \vec{\omega}_o}{2} \right)^2 \times (\beta^2 - 1) + 1 \right)^2}$$

$$S(\vec{\omega}_i, \vec{\omega}_r, \theta_i, \theta_o, \varphi, h, \eta, \beta_L, \beta_A, \mu_a) = \sum_{p=0}^{\infty} S_p(\vec{\omega}_i, \vec{\omega}_r, \theta_i, \theta_o, \varphi, h, \eta, \beta_L, \beta_A, \mu_a)$$

$$S_p(\vec{\omega}_i, \vec{\omega}_r, \theta_i, \theta_o, \varphi, h, \eta, \beta_L, \beta_A, \mu_a) = M_p(\theta_i, \theta_o, \beta_L) \times N_p(\vec{\omega}_i, \vec{\omega}_r, \theta_i, \theta_o, \varphi, h, \eta, \beta_A, \mu_a)$$

$$M_p(\theta_i, \theta_o, \beta_L) = \frac{\text{csch}\left(\frac{1}{\beta^2}\right)}{2\beta_L^2} \times e^{\frac{\sin(-\theta_i) \sin(\theta_o)}{\beta_L^2}} \times I_0\left(\frac{\cos(-\theta_i) \cos(\theta_o)}{\beta_L^2}\right)$$

$$I_0(x) \approx \sum_{i=0}^{10} \left(\frac{x^{2i}}{4^i (i!)^2} \right)$$

(Chiang et al. 2016) : $N_p(\vec{\omega}_i, \vec{\omega}_r, \theta_i, \theta_o, \varphi, h, \eta, \beta_A, \mu_a) = A_p(\vec{\omega}_i, \vec{\omega}_r, \theta_i, \theta_o, \varphi, h, \eta, \mu_a) \times D(\varphi, \beta_A)$

(d'Eon et al. 2011) : $N_p(\vec{\omega}_i, \vec{\omega}_r, \theta_i, \theta_o, \varphi, h, \eta, \beta_A, \mu_a) = \frac{1}{2} \int_{-1}^1 A_p(\vec{\omega}_i, \vec{\omega}_r, \theta_i, \theta_o, \varphi, h, \eta, \mu_a) \times D(\varphi, \beta_A) dh$

$$A_0(\vec{\omega}_i, \vec{\omega}_r, \eta) = \text{Fresnel}\left(\eta, \frac{\arccos(\vec{\omega}_i \cdot \vec{\omega}_r)}{2}\right)$$

$$A_{p \in [1,2]}(\theta_i, \theta_d, h, \eta, \mu_a) = (1 - f)^2 \times f^{p-1} \times T(\mu'_a, h)^p$$

$$A_3(\theta_i, \theta_d, h, \eta, \mu_a) = \sum_{p=3}^{\infty} A_p(\theta_d, h) = \frac{(1 - f)^2 \times f^2 \times T^3}{1 - f \times T}$$

$$f = \text{Fresnel}(\eta, \arccos(\cos(\theta_d) \times \cos(\arcsin(h))))$$

$$\theta_d = \frac{\theta_o - \theta_i}{2}$$

$$\text{Fresnel}(\eta, \psi) = \frac{1 - \eta}{1 + \eta} + \left(1 - \frac{1 - \eta}{1 + \eta}\right) \times (1 - \cos(\psi))^5$$

$$T(\theta_i, \theta_d, h, \eta, \mu_a) = \exp\left(-2\mu'_a \times \left(1 + \cos\left(2 \arcsin\left(\frac{h}{\eta'}\right)\right)\right)\right)$$

$$\eta' = \frac{\sqrt{\eta^2 - \sin^2 \theta_d}}{\cos(\theta_d)}$$

$$\mu'_a = \frac{\mu_a}{\cos(\theta_t)}$$

$$\theta_t = \arcsin\left(\frac{\sin(\theta_i)}{\eta}\right)$$

(Chiang et al. 2016) : $D(\varphi, \beta_A) = l_g(\varphi, \beta_A; -\pi, \pi)$

$$l_g(\varphi, \beta_A; -\pi, \pi) = l_f \left(\varphi, \beta_A \sqrt{\frac{\pi}{8}}; -\pi, \pi \right)$$

$$l_f \left(\varphi, \beta_A \sqrt{\frac{\pi}{8}}; -\pi, \pi \right) = \frac{1}{\frac{1}{1 + \exp(-\frac{\pi}{w})} - \frac{1}{1 + \exp(\frac{\pi}{w})}} \times l_x(\varphi, \beta_A)$$

$$l_x(\varphi, \beta_A) = \frac{e^{-\frac{\varphi}{\beta_A}}}{\beta_A \left(1 + e^{-\frac{\varphi}{\beta_A}} \right)^2}$$

(d'Eon et al. 2011) : $D(\varphi, \beta) = \sum_{k=-\infty}^{\infty} g(\varphi - 2\pi k, \beta)$

$$g(x, y) = \frac{\exp\left(-\frac{x^2}{2y^2}\right)}{\sqrt{2\pi}y}$$

$$S(\varphi_i, \varphi_o, \theta_i, \theta_o, h^-, h^+) = \frac{D(\varphi_i) \cos(\theta_i)^{-2}}{h^+ - h^-} \int_{h^-}^{h^+} S(\varphi_i, \varphi_o, \theta_i, \theta_o, h) \, dh$$

$$S(\varphi_i, \varphi_o, \theta_i, \theta_o, h) = \int \int S(\varphi_i, \theta_i, \Phi, \Theta, h) G(\Theta, \Phi, \theta_o, \varphi_o, B_m, B_n) \cos(\Theta) \, d\Theta \, d\Phi$$

$$G(\Theta, \Phi, \theta_o, \varphi_o, B_m, B_n) = G_M(\Theta - \theta_o | B_m) G_N(\Phi - \varphi_o | B_n)$$

$$S(\varphi_i, \theta_i, \Phi, \Theta, h) = \int_{\Omega} f(x) \, d\mu(x)$$

$$\int_{\Omega} f(x) \, \mathrm{d}\mu(x) \approx \sum_{i=1}^N \vec{\omega}_i f(x_i)$$

$$\int_{\Omega} f(x) \, \mathrm{d}\mu(x) \approx \frac{1}{N} \sum_{i=1}^N \frac{f(x_i)}{p(x_i)}$$

$$\int_{\Omega} f(x) \, \mathrm{d}\mu(x) \approx \frac{1}{N} \sum_{i=1}^N f(x_i)$$

Formule de la plume

$$\begin{aligned} f_s(\vec{\omega}_i, \vec{\omega}_o) = & \left[w_b(\vec{\omega}_o) S_b(\vec{\omega}_i, \vec{\omega}_o, h_b^-(\vec{\omega}_o), h_b^+(\vec{\omega}_o)) G(\vec{\omega}_i) \right. \\ & + w_{bp}(\vec{\omega}_o) S_{bb}(T_{bp}\vec{\omega}_i, T_{bp}\vec{\omega}_o, h_{bp}^-(\vec{\omega}_o), h_{bp}^+(\vec{\omega}_o)) G(T_{bp}\vec{\omega}_i) \\ & + w_{bd}(\vec{\omega}_o) S_{bb}(T_{bd}\vec{\omega}_i, T_{bd}\vec{\omega}_o, h_{bd}^-(\vec{\omega}_o), h_{bd}^+(\vec{\omega}_o)) G(T_{bd}\vec{\omega}_i) \\ & \left. + w_t(\vec{\omega}_o) \delta(1 - \vec{\omega}_i \cdot \vec{\omega}_o) \right] |\vec{\omega}_i \cdot \vec{N}|^{-1} \end{aligned}$$

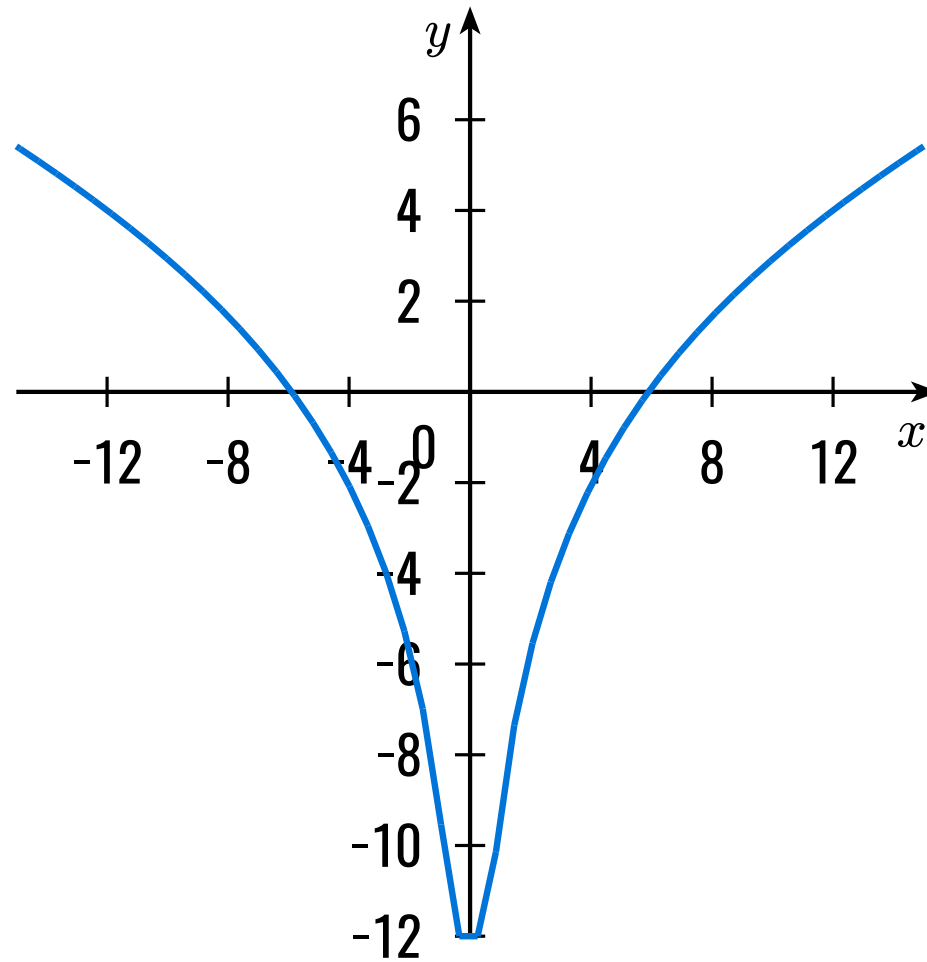
$$p_{t0} = (c_x + a_x \times \cos(\alpha) \quad c_y + a_y \times \cos(\alpha))$$

$$p_{t1} = (c_x + a_x \times \cos(\alpha + \pi) \quad c_y + a_y \times \cos(\alpha + \pi))$$

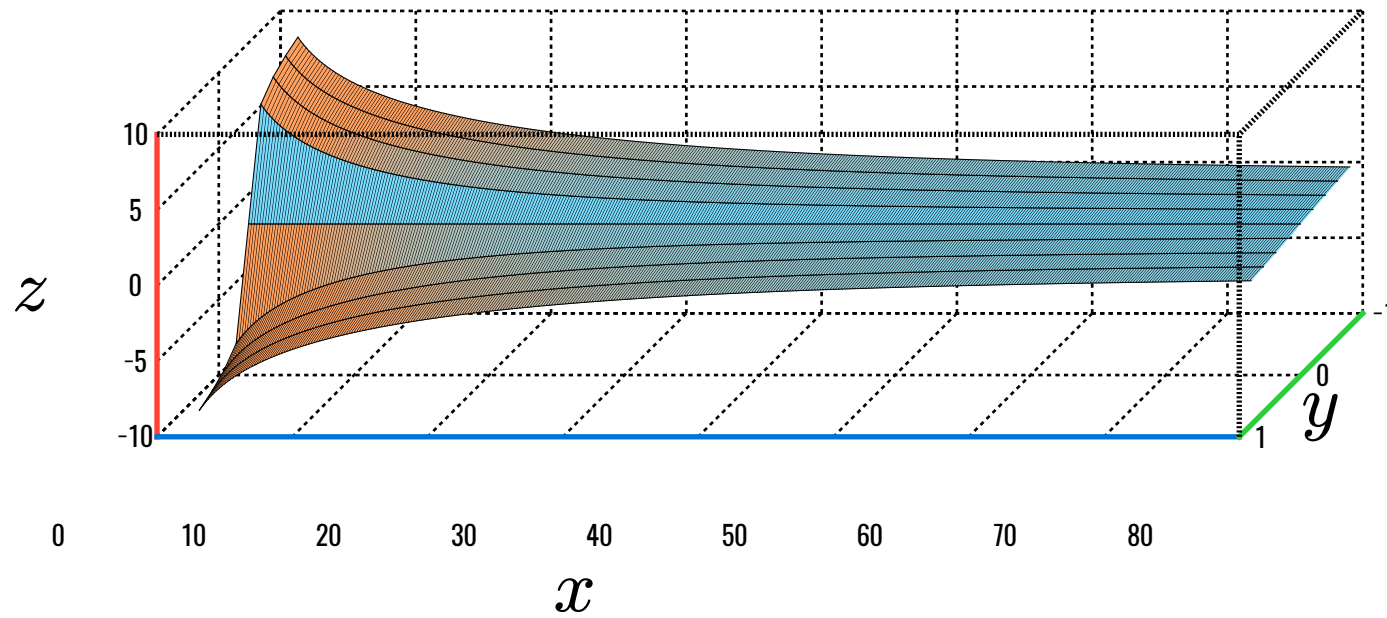
$$\alpha = \tan^{-1} \left(\frac{-a_y \times \omega'_{ox}}{a_x \times \omega'_{oy}} \right)$$

$$A(p_{t0}, p_{t1}) = \omega'_{ox} a_y (\sin(p_{t0}) - \sin(p_{t1})) + \omega'_{oy} a_x (\cos(p_{t0}) - \cos(p_{t1}))$$

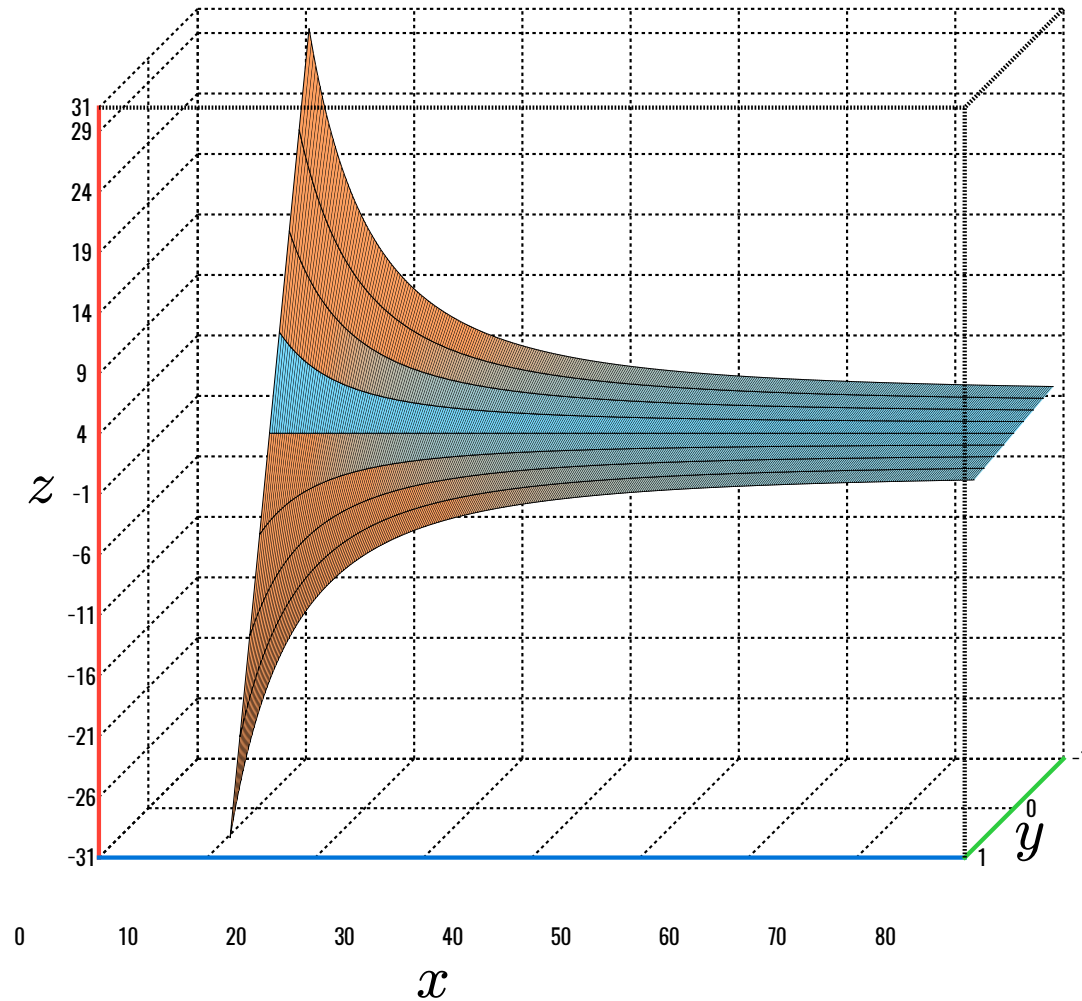
Imprécision de l'implémentation



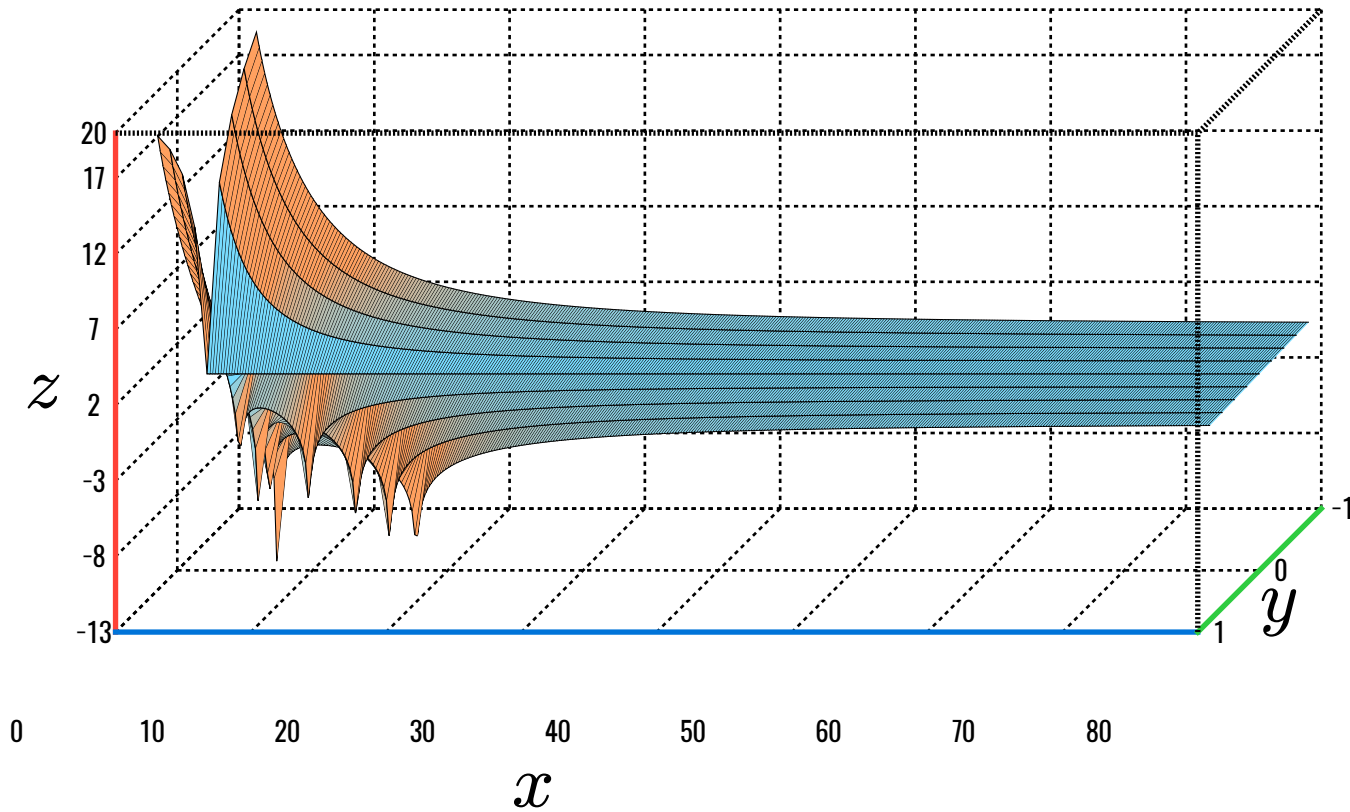
Imprécision de l'implémentation



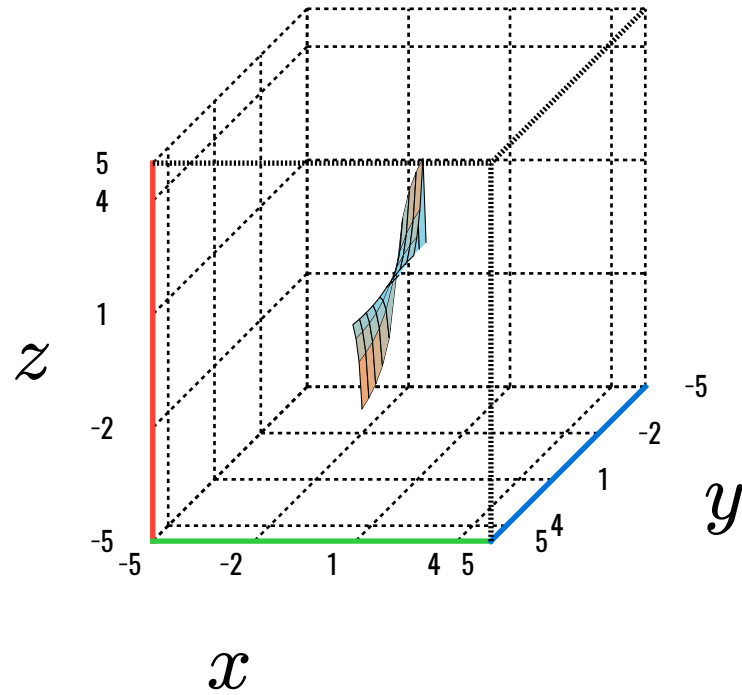
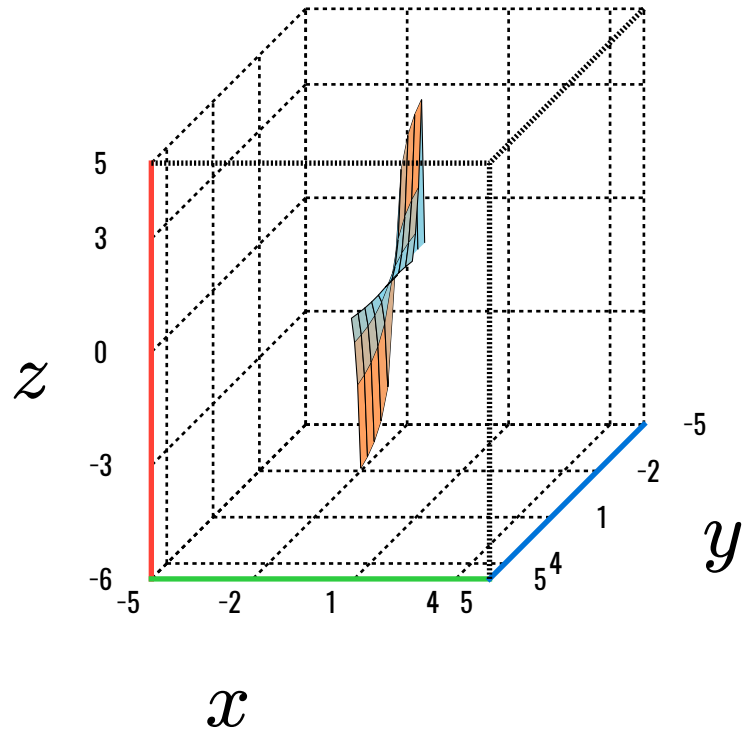
Imprécision de l'implémentation



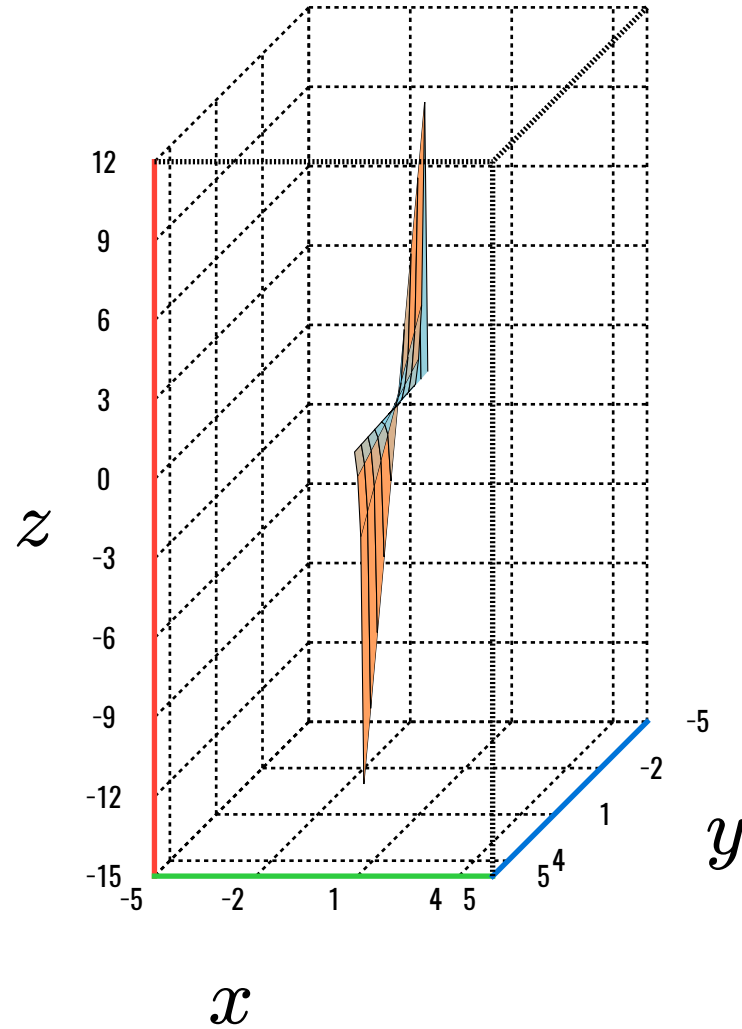
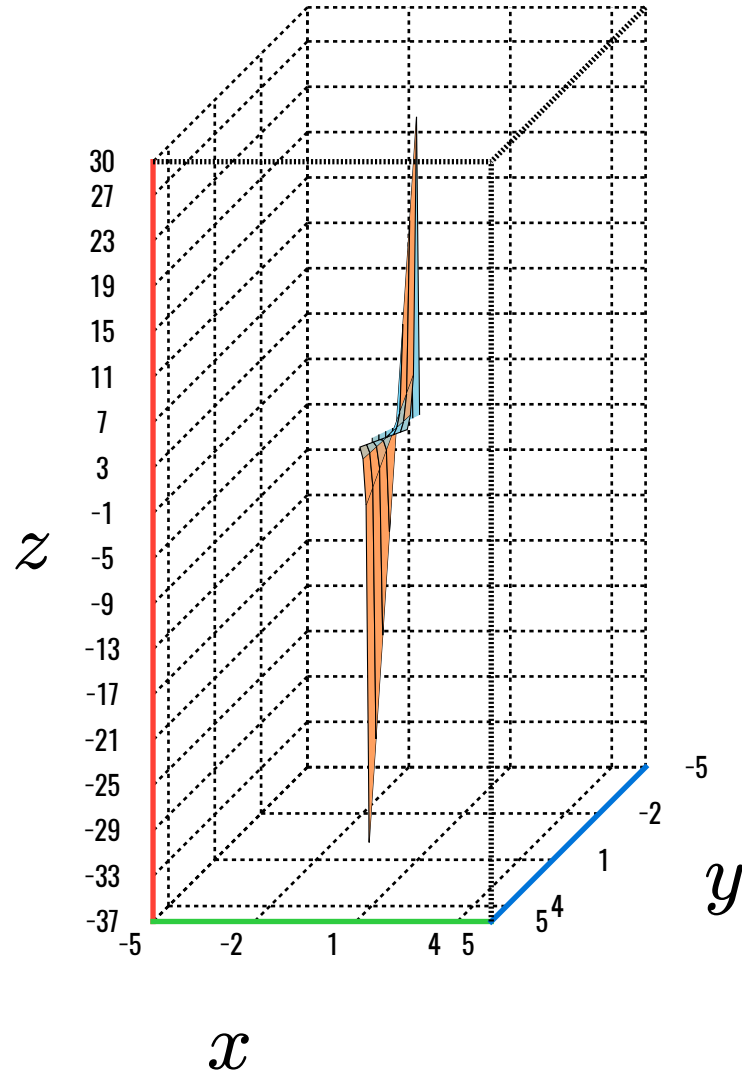
Imprécision de l'implémentation



Imprécision de l'implémentation



Imprécision de l'implémentation



Imprécision de l'implémentation

